MILITARY SPECIFICATION

ANTENNA, AIRBORNE, UHF - BAND,

GENERAL SPECIFICATION FOR

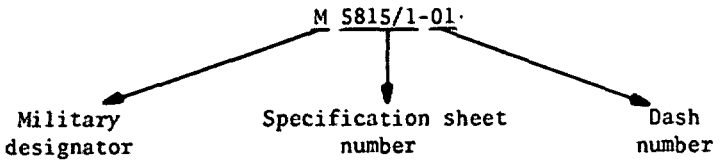
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for ultra high frequency (UHF) band airborne antenna.

1.2 Classification. Antennas covered by this specification are classified by style, as specified (see 3.1).

1.2.1 Military part number. The military part number shall consist of the letter M, the basic number of the specification sheet, and an assigned dash number (see 3.1), as shown in the following:



2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallet, Materials - Handling, Wood (General Construction Requirement).
QQ-S-781	-	Strapping, Steel, Flat and Seals.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape: Packaging, Waterproof.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-E-5400 - Electronic Equipment, Aircraft, General Specification For.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft, General Specification For.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-A-5815/1 - Antenna, Blade, UHF, AT-256A/ARC (Series N Connector).
MIL-C-45662 - Calibration System Requirements.

STANDARDS

FEDERAL

Fed.Std.No. 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-147 - Palletized Unit Loads on 40" x 48" Pallets.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern (see 6.2).

3.2 Qualification. Antenna furnished under this specification shall be a product which has been tested and passed the qualification tests specified in 4.5, and has been listed on or approved for listing on the applicable qualified product list (see 6.2). In addition, the supplier shall obtain certification from the responsible government inspection service that has the quality control requirements of paragraph 4.5.

3.3 General specification. The requirements of MIL-E-5400 for the specified (see 3.1), class of equipment apply as requirements of this specification with the exceptions and additions called out herein. In the event of conflict between this specification and the applicable documents (see 2.1 and 2.2), the requirements of this specification shall govern.

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3.4 Material and finish. Material shall be as specified herein (see 3.1). Where a definite material is not specified, materials and finish used on construction of the antenna shall resist erosion of surfaces due to action of rain, ice crystals and other impinging particles at velocities as specified (see 3.1). Acceptance or approval of any constituent material shall not be construed as a guaranty of acceptance of the finished product.

3.4.1 Nonflammable material. Material shall be nonflammable when tested in accordance with requirement 3 of MIL-STD-454.

3.4.2 Fungus. All materials shall be inert or fungus resistant in accordance with requirement 4 of MIL-STD-454.

3.4.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be in intimate contact.

3.5 Design and construction. Antenna shall be of the design, construction, and physical dimensions specified (see 3.1).

3.5.1 Connectors. The connectors shall be as specified (see 3.1). The material and gaging for the connectors shall conform to the requirements of the applicable specification.

3.5.1.1 Connector caps. All connectors shall be sealed with push-on plastic caps to prevent both damage and the entrance of moisture and foreign material during storage.

3.5.2 Screw threads. Screw threads shall be in accordance with Handbook H28.

3.5.3 Radiation power. The antenna shall be capable of handling the specified power (see 3.1), when subjected to any combination of environmental condition specified herein.

3.5.4 Weight. The weight shall not exceed the limits specified (see 3.1).

3.6 Frequency range. The antenna shall provide performance over the specified frequency range (see 3.1).

3.7 Impedance. When antennas are tested as specified in 4.7.2, the impedance at the connection of the transmission line shall be such that the voltage standing wave ratio (VSWR) produced on the specified RF cable (see 3.1), does not exceed the specified value (see 3.1).

3.8 Sampling probe impedance (when specified). When antennas are tested as specified in 4.7.3, the probe impedance shall be as specified (see 3.1).

3.9 Sampling probe attenuation measurements (when specified). When antennas are tested as specified in 4.7.4, the probe attenuation shall be as specified (see 3.1).

3.10 Radiation pattern. When antennas are tested as specified in 4.7.5, the region extending completely around the antenna, in azimuth, shall, at each zenith angle, be essentially omnidirectional with no nulls greater than 2 dB down from the maximum. The gain at all zenith angles shall be no more than 1 dB below the gain of a matched quarter-wave stub antenna in the same location.

3.11 Polarization. When antennas are tested as specified in 4.7.6, the ratio of vertical or horizontal polarization of signals shall be equal to that obtained from a vertically polarized quarter-wave stub antenna at any frequency specified herein.

3.12 Static load. When antennas are tested as specified in 4.7.7, the antenna shall be capable of withstanding the specific static load (see 3.1).

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3.13 Seal. When tested as specified in 4.7.8, the seal of the antenna shall be capable of withstanding the specified simulated altitude (see 3.1).

3.14 Resistance to solvents. When tested as specified in 4.7.9, antennas shall withstand direct contact with aromatic fuels and hydraulic liquid without causing electrical or mechanical deterioration.

3.15 High temperature exposure. After the high temperature exposure test specified in 4.7.10, the antenna shall show no physical damage, and the impedance shall not exceed the value specified (see 3.1).

3.16 Temperature-Altitude. After the temperature-altitude test specified in 4.7.11.1, the impedance and sampling probe impedance shall not exceed the value specified specified (see 3.1).

3.17 Leakage (immersion). After the leakage test specified in 4.7.11.2, the impedance and sampling probe impedance shall not exceed the value specified (see 3.1).

3.18 Shock. After the shock test specified in 4.7.11.3, the impedance and sampling probe impedance shall not exceed the value specified (see 3.1).

3.19 Vibration. After the vibration test specified in 4.7.11.4, the impedance and sampling probe impedance shall not exceed the value specified (see 3.1).

3.20 Dust. After the dust test specified in 4.7.11.5, the impedance and sampling probe impedance shall not exceed the value specified (see 3.1).

3.21 Humidity. After the humidity test specified in 4.7.11.6, the impedance and sampling probe impedance shall not exceed the value specified (see 3.1).

3.22 Salt spray. After the salt spray test specified in 4.7.11.7, the antenna shall show no physical damage, and the impedance shall not exceed the value specified (see 3.1).

3.23 Marking. Antennas shall be marked in accordance with MIL-STD-130, as specified (see 3.1), and will include the part number, contract, serial number, date code (year, month) and manufacturer's code.

Example: P/N M5815/1-XX
Type AA-BBB (when applicable)
Contract DSA XXXXXXX
Serial No. 4321
75-11 43215

3.24 Workmanship. Antenna shall be manufactured and processed in such a manner as to be uniform in quality, and the shell and mount of the antenna shall be free from tool marks, burrs, deep scratches, and other defects that will affect life, serviceability, aerodynamic drag, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and

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inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) Qualification inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the antennas, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-810.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3), on sample units produced with equipment and procedures normally used in production.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Nonflammable	3.4.1	MIL-STD-454
Fungus	3.4.2	MIL-STD-454
Dissimilar metals	3.4.3	MIL-STD-889

4.5.1 Sample size. Four antennas shall be subjected to qualification inspection. If an antenna produced by a manufacturer is tested and passes qualification tests, the qualification will apply to all antennas produced by the manufacturer with the same manufacturing processes and having the same basic design, except antennas with different mount shall be qualified separately.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspection of group I. The sample units shall then be divided into two groups of 2 units each. The sample units shall then be subjected only to the inspection indicated for that particular group. All units shall then be subjected to group IV.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.5.4 Retention of qualification. To retain qualification, the supplier shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery (Group A), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) The results of tests performed for qualification verification inspection (Group B), including the number and mode of failures. The test report shall include results of all qualification verification inspection tests performed on completed lots during the 12-month period. If the

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test results indicate nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time that the inspection data indicates noncompliance of the product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each type to testing in accordance with the qualification inspection requirements.

TABLE II. Qualification inspection.

Examination or test	Requirement paragraph	Method paragraph
Group I		
Visual and mechanical examination	3.1, 3.3, 3.4 and 3.5	4.7.1
Marking	3.23	4.7.1
Workmanship	3.24	4.7.1
Impedance	3.7	4.7.2
Sampling probe impedance 1/	3.8	4.7.3
Sampling probe attenuation 1/	3.9	4.7.4
Group II		
High temperature exposure 1/	3.15	4.7.10
Temperature - Altitude	3.16	4.7.11.1
Leakage (Immersion)	3.17	4.7.11.2
Shock	3.18	4.7.11.3
Vibration	3.19	4.7.11.4
Dust	3.20	4.7.11.5
Humidity	3.21	4.7.11.6
Salt spray 1/	3.22	4.7.11.7
Group III		
Radiation pattern	3.10	4.7.5
Polarization	3.11	4.7.6
Static load	3.12	4.7.7
Resistance to solvents	3.14	4.7.9
Group IV		
Seal	3.13	4.7.8

1/ When specified (see 3.1).

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all antennas of the same part number, produced under essentially the same conditions, and offered for inspection at one time.

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4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, in the order shown.

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE III. Group A inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination	3.1, 3.3, 3.4, 3.5	4.7.1	0.65	1.5
Marking	3.23	4.7.1		
Workmanship	3.24	4.7.1		
Impedance	3.7	4.7.2		
Sampling probe impedance 1/	3.8	4.7.3		
Sampling probe attenuation 1/	3.9	4.7.4		
Seal	3.13	4.7.8		

1/ When specified (see 3.1).

4.6.2 Qualification verification inspection. Qualification verification inspection shall consist of group B. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.4), delivery of products which have passed group A shall not be delayed pending the results of these qualification verification inspections.

4.6.2.1 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table IV, in the order shown. Group B inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

4.6.2.1.1 Sampling plan. Four sample units shall be selected every 6 months. Upon passing this inspection, the supplier may select 4 sample units every 12 months. If the second level of sampling is passed two successive times, the supplier may select 4 sample units every 24 months. In the event of a failure, sampling shall revert to the 6-month interval.

4.6.2.1.2 Failure. If one or more sample units fail to pass group B inspection, the sample shall be considered to have failed.

4.6.2.1.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.6.2.1.4 Noncompliance. If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample

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failed, at the option of the Government). Group A inspection may be reinstituted; however, final acceptance shall be withheld until the group B reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

TABLE IV. Group B inspection.

Examination or test 1/	Requirement paragraph	Method paragraph
Radiation pattern	3.10	4.7.5
Polarization	3.11	4.7.6
Static load	3.12	4.7.7
Seal	3.13	4.7.8
Resistance to solvents	3.14	4.7.9
High temperature exposure 1/	3.15	4.7.10
Temperature-Altitude	3.16	4.7.11.1
Leakage (Immersion)	3.17	4.7.11.2
Shock	3.18	4.7.11.3
Vibration	3.19	4.7.11.4
Dust	3.20	4.7.11.5
Humidity	3.21	4.7.11.6
Salt spray	3.22	4.7.11.7

1/ When specified (see 3.1).

4.6.3 Inspection of preparation for delivery. Except when commercial packaging is specified, the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract or purchase order (see 6.2).

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. Antennas shall be examined to verify that materials, design, construction, physical dimensions, finish, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.5, 3.23, and 3.24).

4.7.2 Impedance (see 3.7). Impedance measurements shall be made with the antenna installed flush in the center of an 8-foot square ground plane using a slotted-line method. The slotted line shall be equipped with a tunable RF probe covering the appropriate frequency range. The impedance of the antenna shall be measured and recorded over the frequency range (see 3.1) at interval no greater than 25 MHz. (Sweep frequency techniques may be used.) The cable length between the antenna and the measuring equipment shall be no greater than six feet. Impedance measurements of the antenna shall be made after environmental tests as well as temperature extremes.

4.7.3 Sampling probe impedance (see 3.8). The impedance of the sampling probe shall be measured in accordance with 4.7.2, over the frequency range specified (see 3.1).

4.7.4 Sampling probe attenuation measurements (see 3.9). Attenuation measurements shall be made with the signal source connected through a 50 ohm calibrated RF attenuator to the antenna connector and the output of the sampling probe connected to the calibrated receiving detector. The output of the probe shall be noted on the detector, after which, the antenna and probe shall be removed from the circuit and a suitable

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connector-adapter substitute therefore. Without changing power or sensitivity, the attenuator shall be adjusted to produce the reading noted previously on the detector. The difference in reading, in dB, on the attenuators with and without the antenna and probe in place shall be the probe attenuation. All RF interconnections shall be made by means of nominal 50 ohm coaxial cables (and connectors) and mismatch of both the signal generator and the detector used for making the measurements shall be reduced as low as possible. The attenuation of the antenna shall be measured over the frequency range (see 3.1) at interval no greater than 25 MHz. (Sweep frequency techniques may be used.) The cable length between the antenna and the measuring equipments shall not be greater than 6 feet. Each antenna shall be subjected to the attenuation tests both before and after completion of the environmental tests.

4.7.5 Radiation pattern (see 3.10). Measurements of the radiation pattern shall be made with the antenna mounted flush in the center of an 8-foot-square plane. The pattern shall be made on a continuously recording radio range of the automatic type. Care shall be taken to avoid errors due to reflections from nearby objects, including earth. Patterns shall be made at the horizon (ground plane) and a vertical angle of 30 degrees above the ground plane and to azimuth angles of 30, 60, 90, and 95 degrees. Like patterns shall be made for comparison, using the same test power levels on a matched quarter-wave stub antenna. The above tests shall be made at the specified frequencies (see 3.1), but need not be repeated after environmental tests.

4.7.6 Polarization (see 3.11). Polarization tests shall be conducted in the continuously recording radio range in the same manner as the tests for radiation pattern measurements. Polarization-radiation patterns need be made at only the one specified frequency (see 3.1), before exposure to environmental tests. Polar radiation diagrams shall be made for the following conditions:

(a) For each test sample:

(1) Pattern in azimuth

(a) With transmitting antenna alined the same as the antenna under test.
(b) At the same power level as above with the transmitting antenna cross polarized with respect to the antenna under test.
(c) Same as (b) with increased transmitter power or increased receiver sensitivity by a fixed amount necessary for examination of pattern in detail.

(2) Patterns in the transverse vertical plane as listed under (1) above.

(3) Patterns in the longitudinal vertical plane as listed in (1) above.

(b) For the quarter-wave stud, repeat all measurements listed under "a" at identical power levels and sensitivities.

4.7.7 Static load (see 3.12). The antenna shall be normally mounted on a fixture and a bladder draped on one side similar to figure 1. The bladder shall then be inflated until it is just making contact with the antenna surface. An air pressure reading shall then be taken and the internal air pressure shall then be increased to the specified pressure or to the pressure simulating the load pound (see 3.1) and held for 5 minutes. The pressure shall then be released and the same procedure repeated on the opposite side. This shall be considered as 1 cycle. The test shall be 2 cycles in duration. On completion of the second cycle the antenna shall be moved and carefully examined for deformation and structural failure. Impedance tests shall be made and the VSWR shall be as specified (see 3.1). When specified (see 3.1), the above test shall be applied to the other 2 sides.

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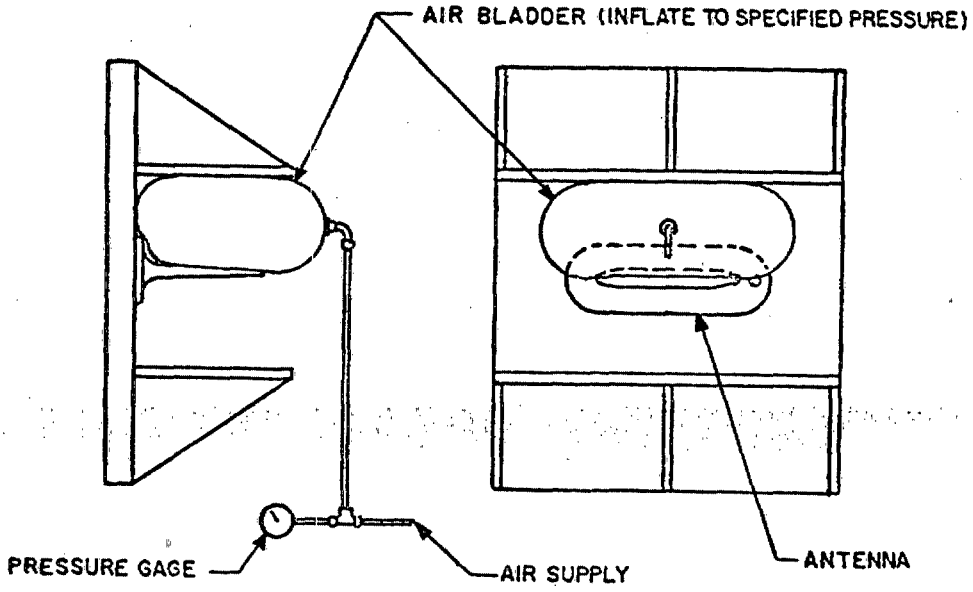


FIGURE 1. Static load test.

4.7.8 Seal (see 3.13). All samples shall be subjected to a seal test by the altitude chamber method. The antennas shall be fully submerged in a water-filled, transparent container. The test antennas may be covered with a mating cap simulating the mating connection. The chamber shall be evacuated to a pressure of 1.32 inches of mercury simulating an altitude of 70,000 feet and remain there for a minimum of 5 minutes. The chamber pressure shall then be raised to 2.5 inches of mercury simulating an altitude of 57,000 feet and maintained at that condition for 10 minutes. The chamber pressure shall then be raised to standard atmospheric pressure for a perioc of 1 hour. The antennas, upon completion of the seal test, shall be removed and all excess moisture wiped from the antenna. Any evidence of leakage either by a flow of bubbles during evacuation or subsequent entrance of water shall be considered a failure Immediately after the examination, the impedance test shall be made and the VSWR shall be as specified (see 3.1).

4.7.9 Resistance to solvents (see 3.14). Place the sample antenna, completely immersed, in the test liquid for 24 hours. Upon removal, shake or wipe excess surface liquid from the antenna and the impedance measured in accordance with 4.7.2 and 4.7.3. The above procedure shall be performed with each of the following test liquids: JP-4 fuel (MIL-T-5624) and hydraulic fluid (MIL-H-5606).

4.7.10 High temperature exposure (see 3.15). The sample antennas shall be placed within a chamber and the internal temperature shall be raised to 220°C (428°F). The antenna shall remain in this environment for 2 hours; at the end of this time, the antenna shall be removed from the chamber and immediately measured for impedance.

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4.7.11 Environmental tests (see 3.6). Environmental tests shall be in accordance with MIL-STD-810, in the sequence shown herein. Following each test, the antennas must meet the requirements of 3.7 and 3.8.

4.7.11.1 Temperature-Altitude (see 3.16). Method 504.1.

(a) Procedure I.
(b) Equipment catalogue 7.

4.7.11.2 Leakage (Immersion) (see 3.17). Method 512.1.

(a) Procedure III.
(b) Liquid: Water.
(c) Pressure: 1 and 2.5 inches of mercury.

4.7.11.3 Shock (see 3.18). Method 516.

(a) Procedure I.
(b) Half sine wave, 15 g's peak, 11 ms duration.
(c) Operation not required during test.

4.7.11.4 Vibration (see 3.19). Method 514.2 - II.

(a) Procedure I.
(b) Curve E (15g) 5-2,000 Hz.

4.7.11.5 Dust (see 3.20). Method 510.1.

4.7.11.6 Humidity (see 3.21). Method 507.

(a) Procedure I.

4.7.11.7 Salt spray (see 3.22). Method 509, procedure I.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, or as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Antennas shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Antennas shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Each antenna shall be individually unit packaged in accordance with MIL-P-116, submethod IC-2, insuring compliance with the applicable requirements of that specification. The container shall conform to PPP-B-566, PPP-B-676, or PPP-B-636.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Antennas shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

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5.2 Packing. Packing shall be level A, B, or C, or as specified (see 6.2).

5.2.1 Level A. The packaged antennas shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged antennas shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged antennas shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Antennas, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Antennas, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1, except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Antennas, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size, and kind commonly used for this purpose. These unitized loads shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package, exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.3.

5.4.3 Army procurements.

5.4.3.1 Level A (maximum military protection) and level B (minimum military protection) packing. For level A packing, the fiberboard containers shall not be banded, but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4, or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification, except that metal

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strapping shall conform to QQ-S-781, type I, finish A. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more, and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.3.2 Levels A and B unitization. For levels A and B unitization, softwood pallets conforming to NN-P-71, type IV, size 2 shall be used. Weather resistant fiberboard caps shall also be used for level B unitization. The loads for both levels shall be bonded to the pallets by strapping conforming to QQ-S-781, type I, finish A or shrink film (see 5.2.4.1 and 5.2.4.2).

5.4.3.3 Commercial packaging. Commercial packaging (including unit packaging, packing, and marking) shall be in accordance with Federal Standard No. 356.

6. NOTES

6.1 Intended use. The antenna is a radiation element having an essentially omnidirectional vertically polarized radiation pattern intended for use with airborne communication.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Title, number and date of the applicable specification sheet and the complete part number (see 1.2.1 and 3.1).
(c) Inspection of commercial packaging (see 4.6.3).
(d) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(e) Special marking, if required (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Air Force (Code 85); information pertaining to qualification of products may be obtained from Engineering and Technology Support Division, Interservice/Interagency Support Office, AFLC/LOIE, Wright-Patterson Air Force Base, Ohio 45433. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification".

6.3.1 Design activity. All changes and waivers which could modify the antennas in any way from the articles qualified, must be approved by the AFLC/LOIE, Wright-Patterson Air Force Base, Ohio 45433.

6.3.2 Copies of "Provisions Governing Qualifications" SD-6 may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

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6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Air Force - 85

Review activities:
Army - AV
Air Force - 11, 99
DSA - ES

Preparing activity:
Air Force - 85

(Project 5985-0808)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-A-5815E Antenna, Airborne, UHF - Band, Gen Spec for FSC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER 5985

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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